

WELCOME TO THIS CASE STUDY!

VESPERA SPACES

Integrated Business Management System

Edmund Loke

Buckle up — we've got a lot to show you today

→ **Project Overview**

User Research

The Design

Refinements

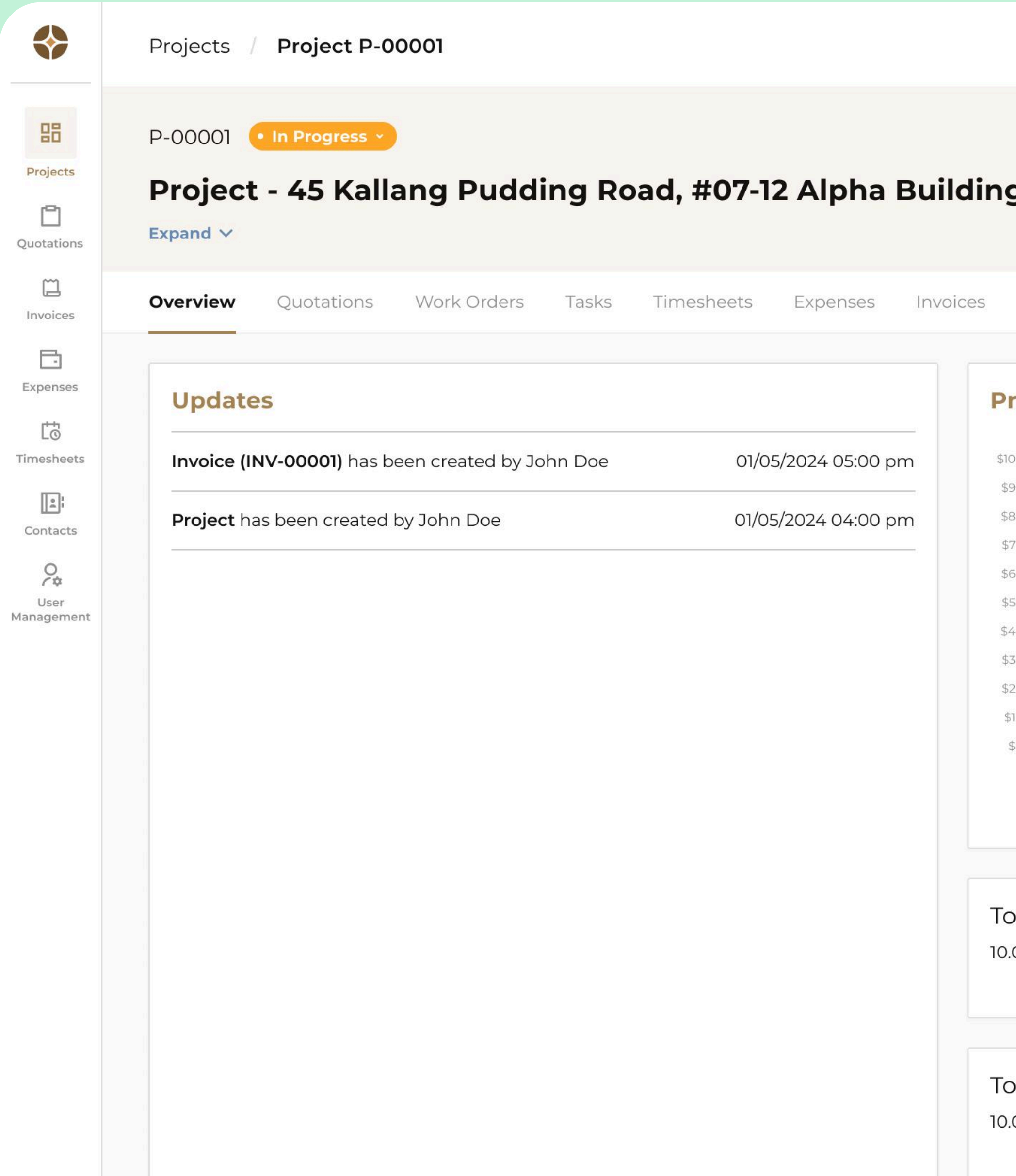
Lessons learned

PROJECT OVERVIEW

Introduction

This is a project for an interior design company where they would like a system that can integrate their processes to one system such as managing quotations, project tracking , task management and expenses/invoice tracking.

By combining all this function into one system, it allows a unified experience where each modules and functions are integrated closely within each other.



PROJECT OVERVIEW

About the Project



The Problem

Currently, the business uses separate systems for different processes, creating inefficiencies. For example, employees must manually switch between systems for tasks like matching invoices to projects.



The Goal

This system is designed to eliminate workflow barriers by seamlessly integrating all business processes. By automating manual tasks, it enhances operational efficiency while minimizing human error.

Project Overview

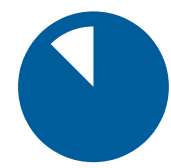
→ **User Research**

The Design

Refinements

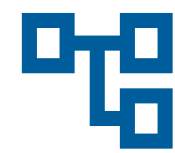
Lessons learned

Pain Points



Time Consuming

Users are required to manually cross-reference invoices and project records to track expenses resulting in an increase of time required to cross check and prone to errors.



Inefficient Tracking

Unable to efficiently keep track of which projects the users are currently working on or the time they have spent on each project or tasks.



Frustrations

No systematic way to store or retrieve historical quotations, work orders efficiently or expenses resulting in higher chance of confusion when looking back at past documents.

Problem Statement

Joe is a project manager who needs to an unified platform to manage projects and user’s timesheets easily because he wants to focus on strategic priorities instead of administrative processes.

Joe

Age: 32
Hometown: Kuala Lumpur
Occupation: Project Manager
Education: Master of Business



Summary

Joe is an experienced Project Manager with over five years in interior design, Joe excels at team coordination and client negotiations. Yet his effectiveness is undermined by fragmented systems he must re-enter project details just to log hours in the timesheet system. In addition, whenever he starts a project he will need to enter the details again to the system to assign tasks to the team. This duplication of effort creates operational delays and frustrates his team.

Goals

- Have an efficient and customized solution to reduce duplication of steps and redundancy
- Seeks a streamlined solution to review and approve team timesheets.

Pain Points

- Difficulty in creating and keeping track of projects
- Having to enter project data twice, first for tracking, then for timesheets



“I am deeply passionate about continuous learning and exploration, which broadens my perspective and enhances my adaptability.”

Problem Statement

Lilian is a director who needs a platform to easily manage and approve expenses because she wants to speed up business processes and avoid unnecessary delays.

Lilian

Age: 48
Hometown: Kuala Lumpur
Occupation: Director
Education: Master of Business



Summary

Lilian is a director in the company and is a person that values efficiency. However, she faces hurdles in the business processes varying from unclear project expense tracking to missing approval status visibility. This disorganization delays workflows, often leaving project managers unaware of approved expenditures creating bottlenecks in financial oversight and team communication.

Goals

- Ⓜ To have a way to easily approve expenses and update the team immediately.
- Ⓜ Maintains complete historical expense records with full visibility into past expenditures.

Pain Points

- Ⓜ Time consuming when there is a need to view past expenses
- Ⓜ Delayed workflows due to lack of updates of approved or declined expenses



“Efficiency and hard work drive me and I see them as the foundation of achievement.”

Project Overview

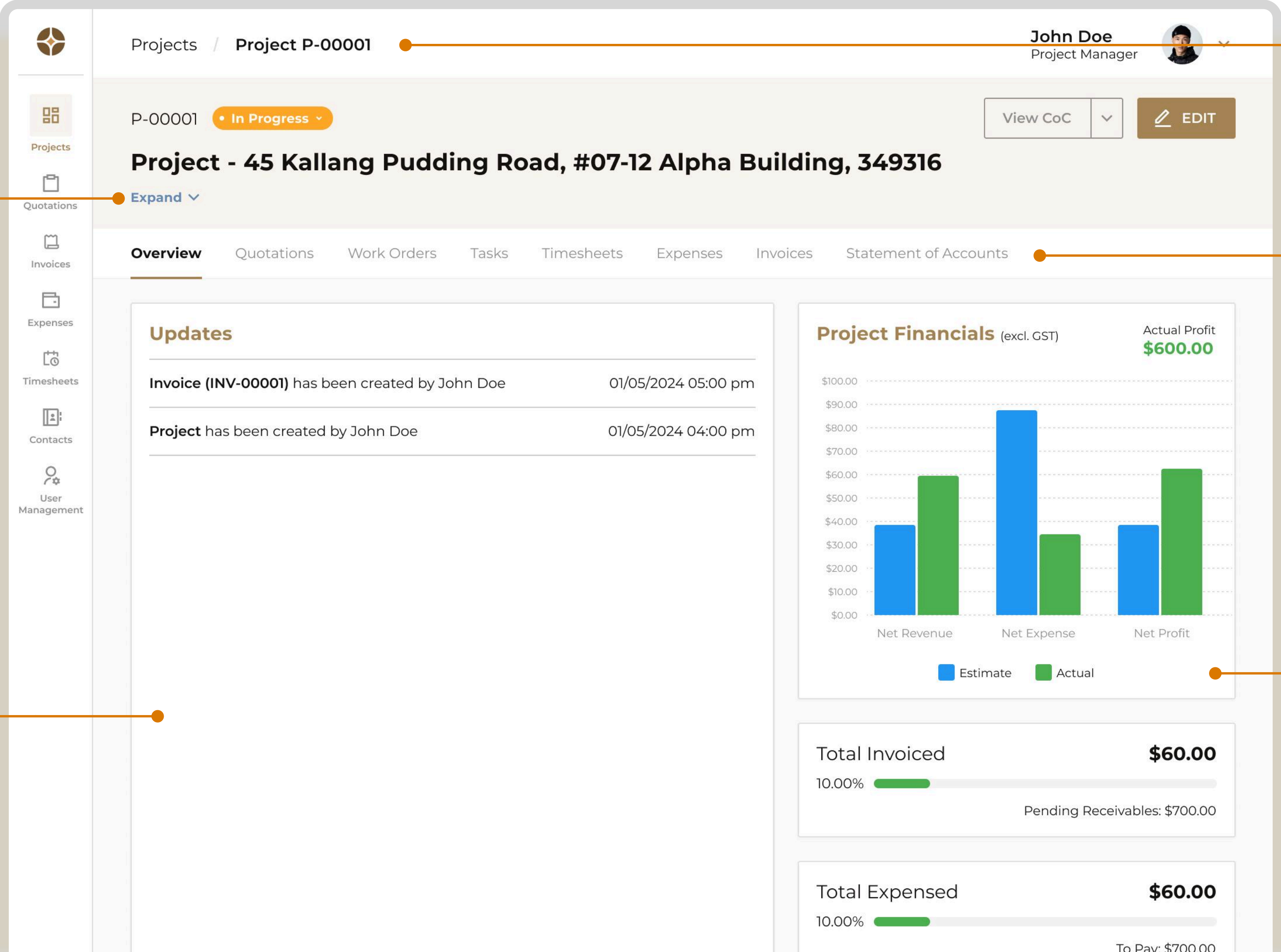
User Research

→ **The Design**

Refinements

Lessons learned

Project Module



To reduce visual clutter, project details are hidden by default and can be revealed using the expand button.

Allow users to have a detailed view on the project updates.

Breadcrumbs are included to enable seamless navigation between pages.

Enables users to access project-specific modules directly, without the need to navigate the main modules.

Providing an overview of financial information about the project such as expense spent and amount received.

Project Module

To reduce visual clutter, project details are hidden by default and can be revealed using the expand button.



Projects

Quotations

Invoices

Expenses

Timesheets

Contacts

User Management

Projects / Project P-00001

John Doe

Project Manager



P-00001 In Progress

View CoC



 EDIT

Project - 45 Kallang Pudding Road, #07-12 Alpha Building, 349316

Collapse 

Project Details

Quotation	PO Number
Q-00001 - Project Development H	PO-00001
Manager in Charge	Currency
Adam Lim	SGD
Start Date	Due Date
01/05/2024	30/04/2024
Remark	
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris accumsan facilisis mi at sodales. Integer congue mauris a mattis dictum. Nulla ac sem sit amet augue aliquet convallis ut ac odio. Aliquam viverra mi ut erat accumsan cursus. Cras id sollicitudin nisl. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet. Cras id sollicitudin nisl. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet...	
Created On	Created By
01/05/2024	John Doe

Project Site Details

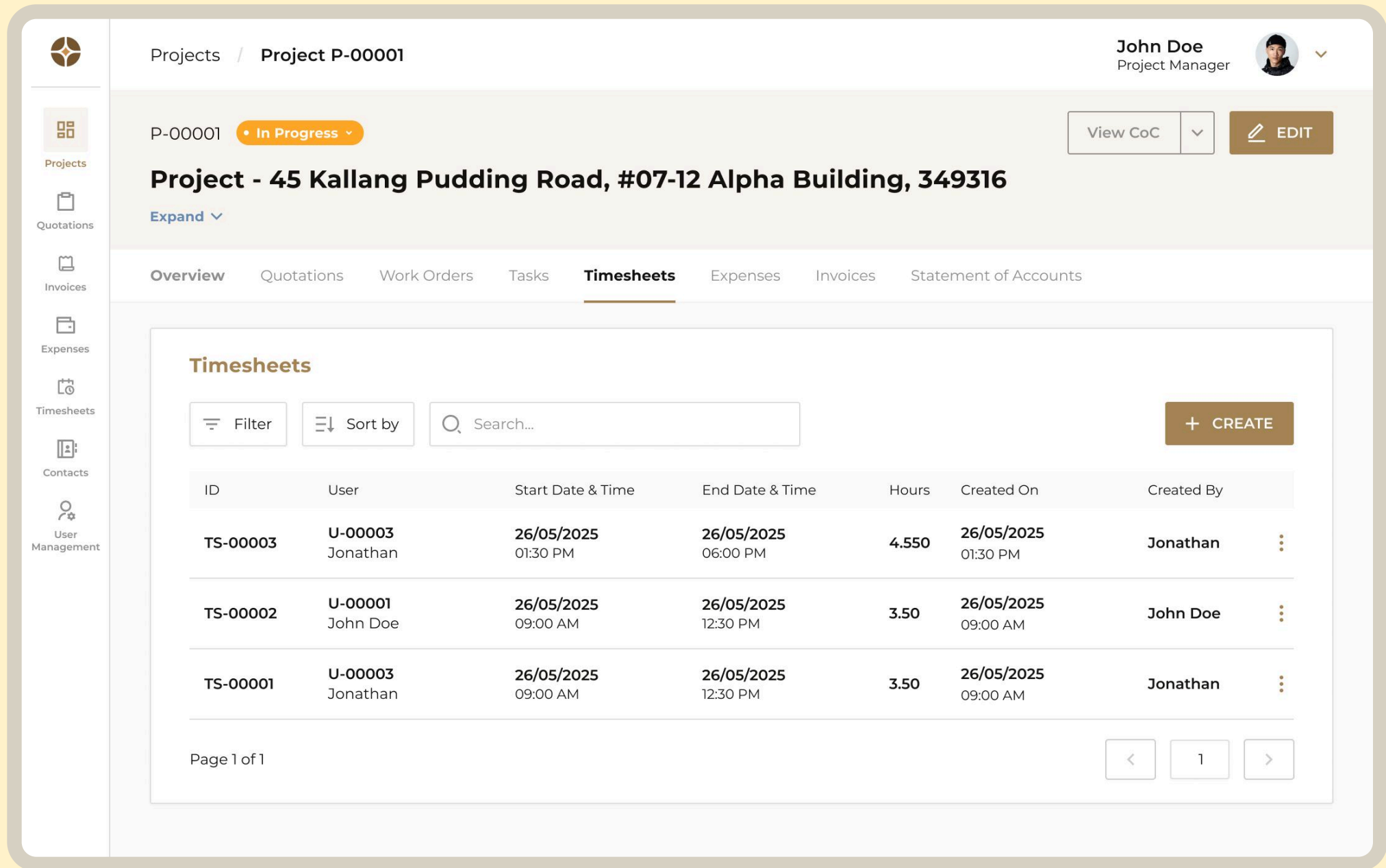
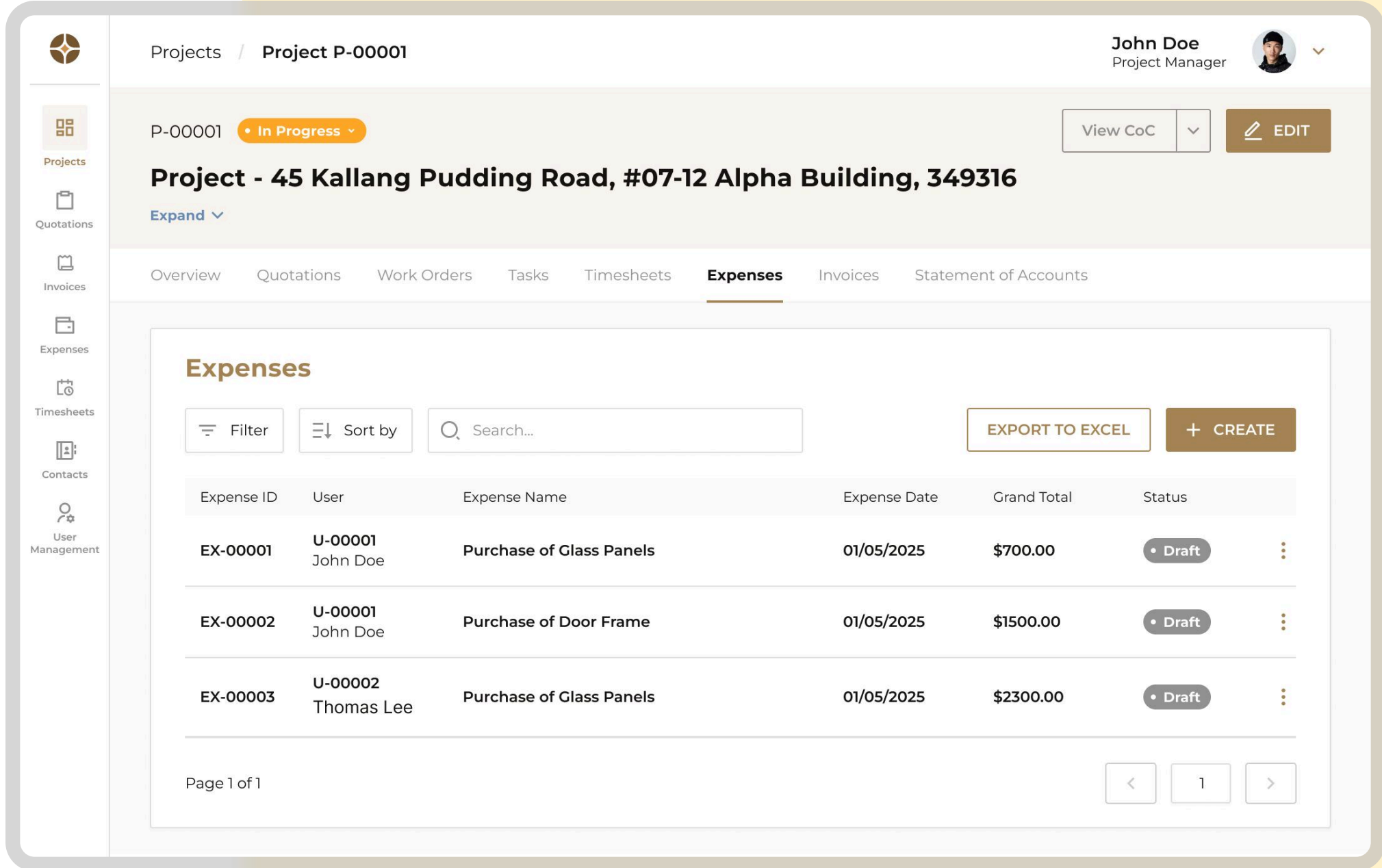
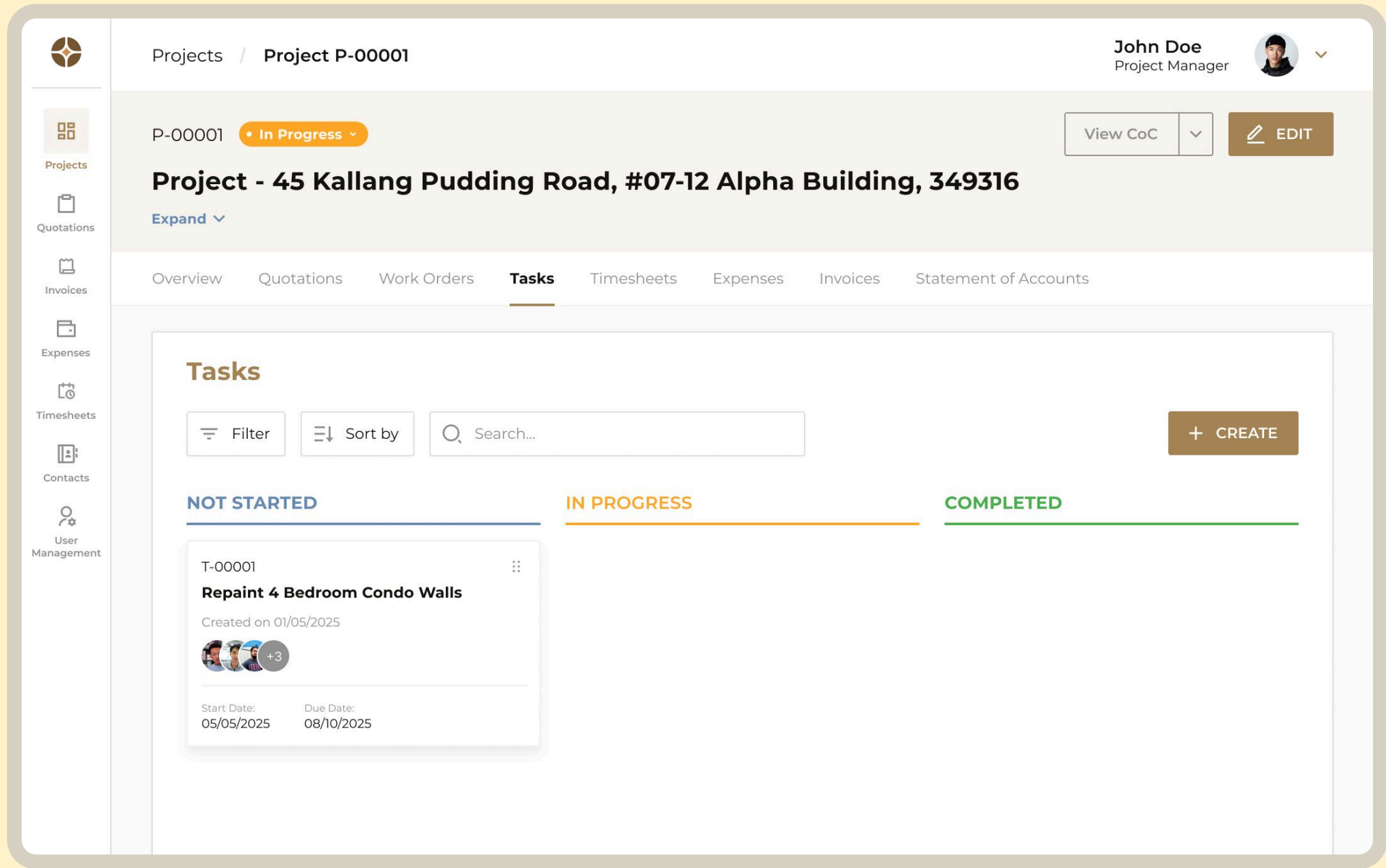
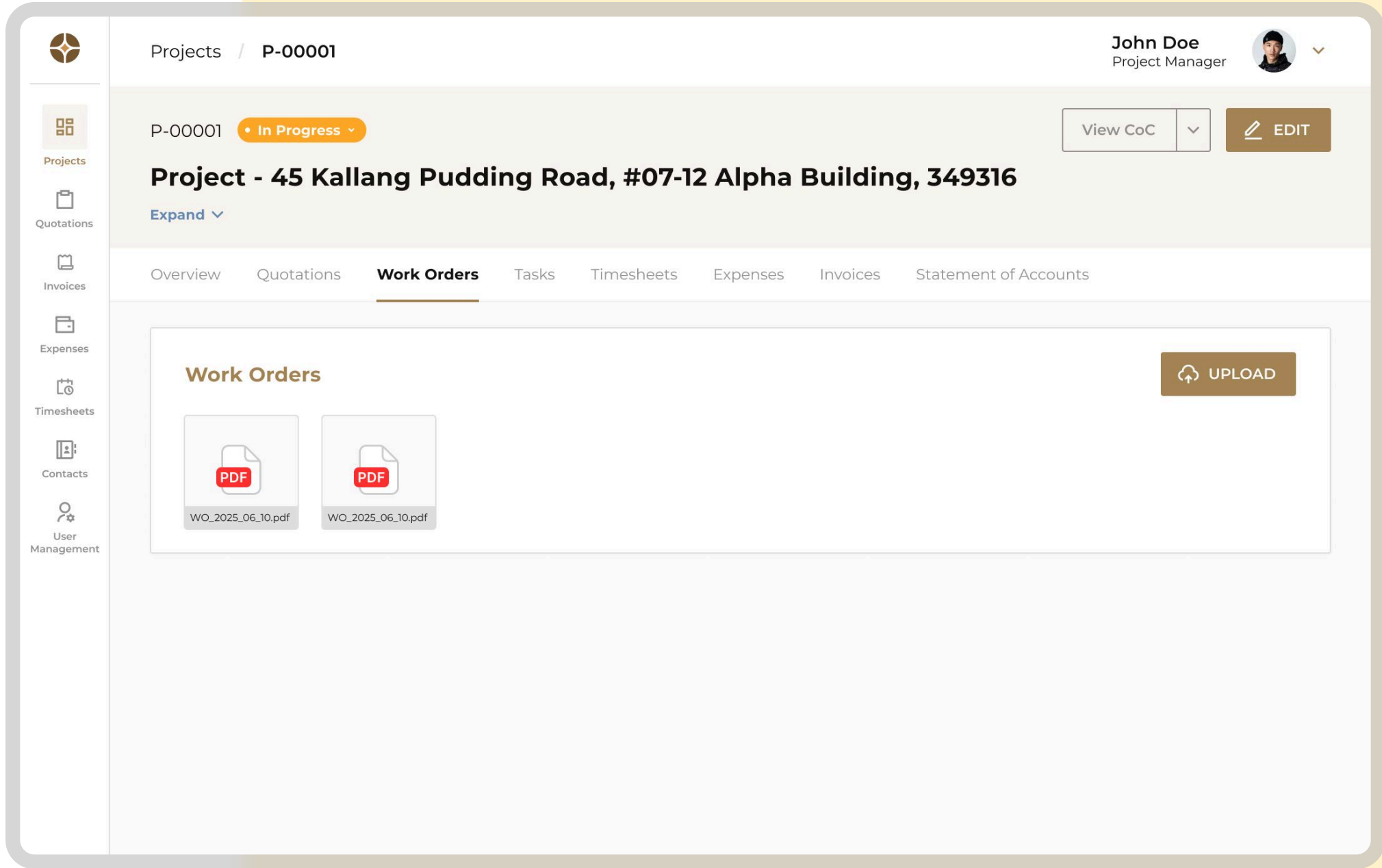
Project Site	Project Address
Project H	123 Raffles Place, 22-0188, Business Park S(048621)

Client Details

Client Name	Client Address
Project H	123 Raffles Place, 22-0188, Business Park S(048621)

Breadcrumbs are included to enable seamless navigation between pages.

Project Module



Timesheets

Project Module

Projects / Project P-

Projects

Quotations

Invoices

Expenses

Timesheets

Contacts

User Management

P-00001

In Progress

Project - 45 Kalla

Expand

Overview

Quotations

Timesheets

Filter

Sort

ID	User
TS-00003	U-000 Jonatr
TS-00002	U-000 John C
TS-00001	U-000 Jonatr

Page 1 of 1

Timesheet TS-00004

DELETE

SAVE & CLOSE

Created On 30/04/2025 09:00 AM

Created By John Doe

Project

P-00001 - 45 Kallang Pudding Road, #07-12 Alpha Bui...

User

U-00001 - John Doe

Start Date & Time

27/05/2025 09:00 AM

End Date & Time

27/05/2025 12:30 PM

Description (Optional)

Performed intricate paintwork, including surface preparation and applying multiple coats. Ensured all areas were evenly covered and conducted thorough inspections to guarantee a flawless finish.

Hours

3.50

Close drawer without saving any changes made.

Autofill when timesheet created from project module

Timesheet Module

Timesheets

Projects

Quotations

Invoices

Expenses

Timesheets

Contacts

User Management

Filter

Sort by

Timesheet TS-00001

DELETE

SAVE & CLOSE

Created On 30/04/2025 09:00 AM

Created By John Doe

Project

P-00001 - 45 Kallang Pudding Road, #07-12 Alpha Bui...

User

U-00001 - John Doe

Start Date & Time

27/05/2025 09:00 AM

End Date & Time

27/05/2025 12:30 PM

Description (Optional)

Performed intricate paintwork, including surface preparation and applying multiple coats. Ensured all areas were evenly covered and conducted thorough inspections to guarantee a flawless finish.

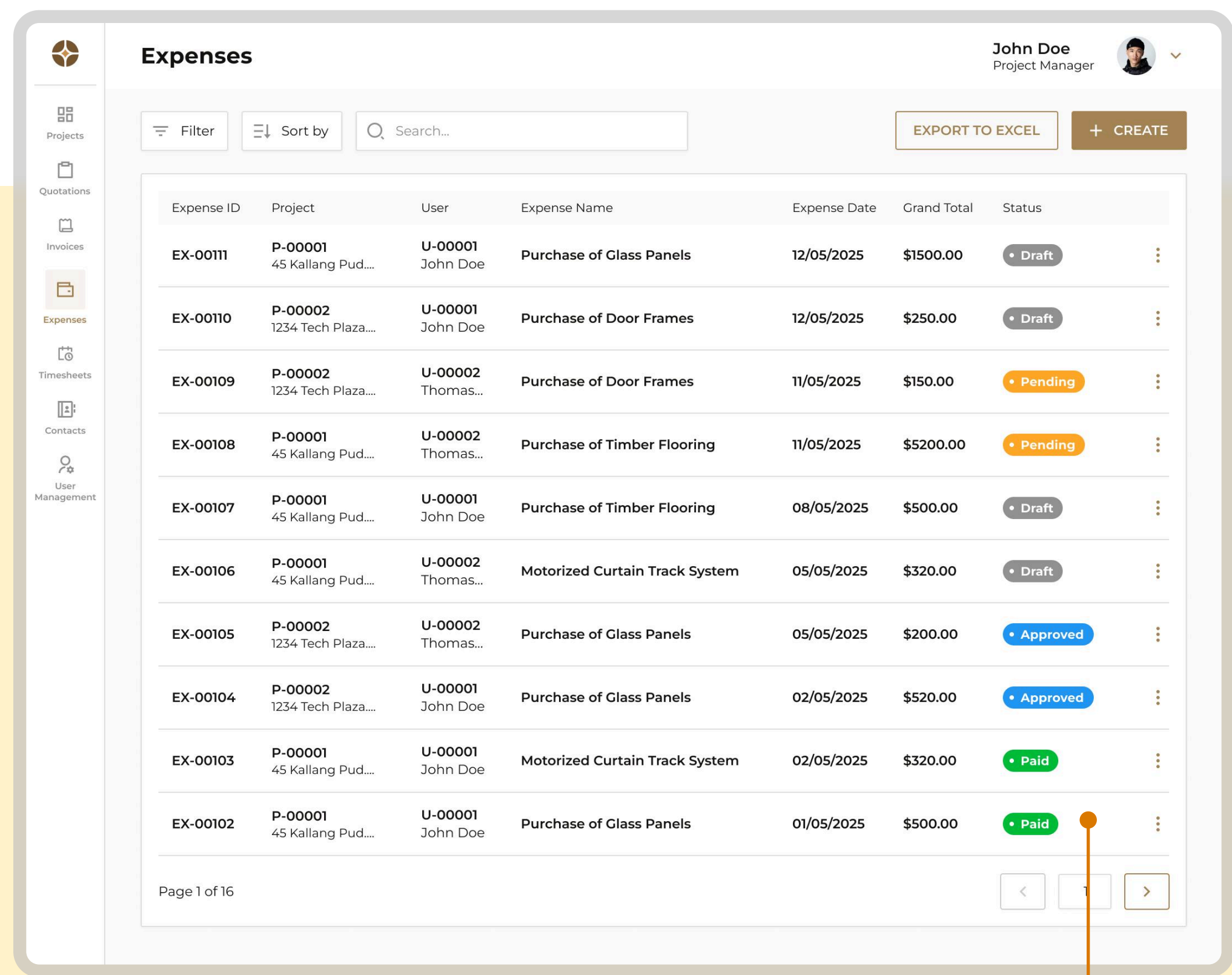
Hours

3.50

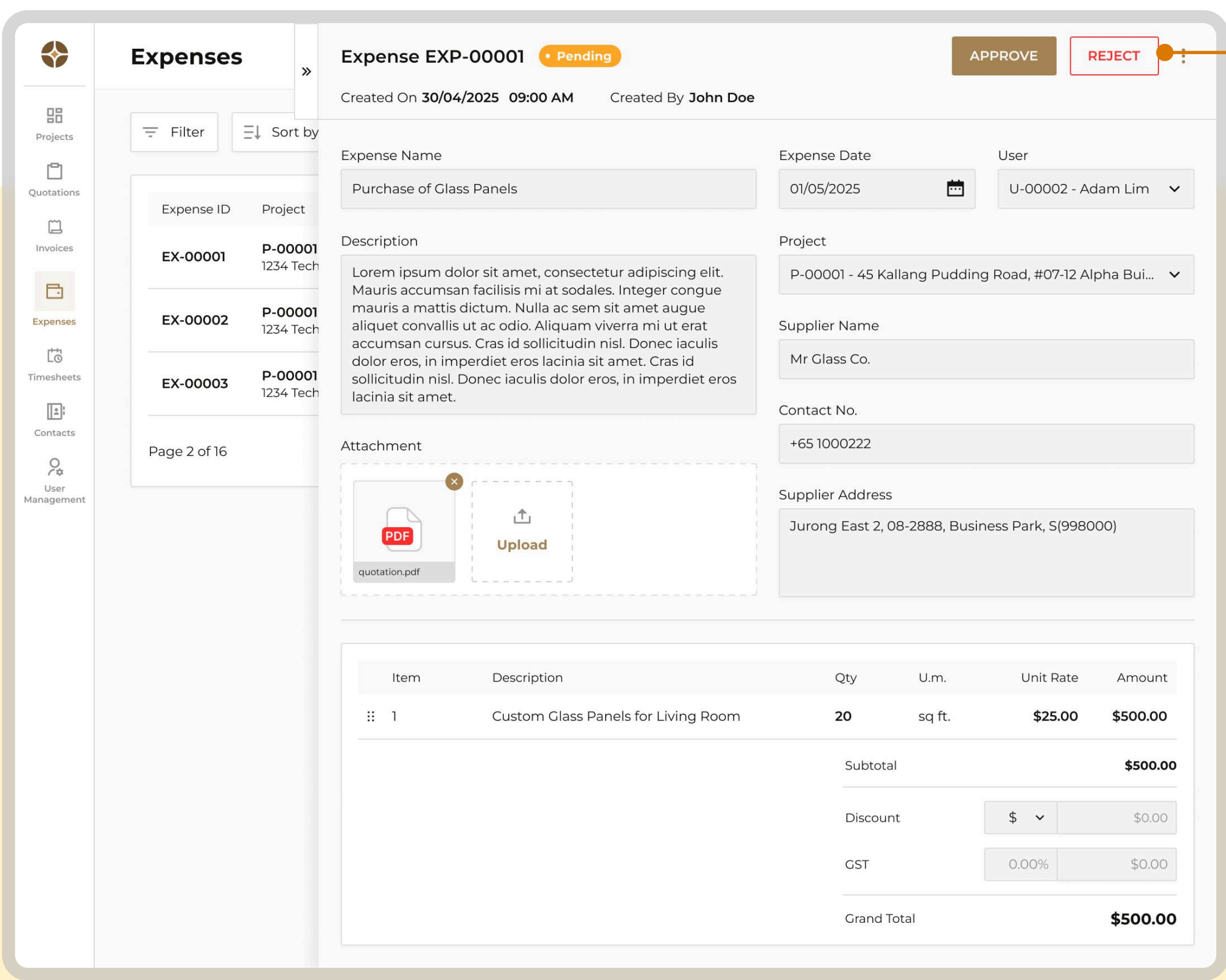
The drawer layout displays previous screen context, helping users maintain their spatial orientation.

Timesheet creation details are position at the top for clear visibility and hierarchy

Expenses Module - Admin



Inclusion of different statuses to allow easy tracking of approved or rejected expenses.



Key CTAs are positioned top-right to remain consistency, reducing cognitive load and improving action speed.

Expenses Module - User

A confirmation dialog will appear upon submission

Expenses

Expense EXP-00001

Draft

Created On 30/04/2025 09:00 AM

Created By John Doe

Filter

Sort by

Expense Name

Purchase of Glass Panels

Expense Date

01/05/2025

User

U-00002 - Adam Lim

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris accumsan facilisis mi at sodales. Integer congue mauris a mattis dictum. Nulla ac sem sit amet augue aliquet convallis ut ac odio. Aliquam viverra mi ut erat accumsan cursus. Cras id sollicitudin nisl. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet. Cras id sollicitudin nisl. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet.

Project

P-00001 - 45 Kallang Pudding Road, #07-12 Alpha Bui...

Supplier Name

Mr Glass Co.

Contact No.

+65 1000222

Attachment

quotation.pdf

Upload

Supplier Address

Jurong East 2, 08-2888, Business Park, S(998000)

☐	Item	Description	Qty	U.m.	Unit Rate	Amount	
☐	1	Custom Glass Panels for Living Room	20	sq ft.	\$25.00	\$500.00	
+ Add New Line							
Subtotal						\$500.00	
Discount					\$	\$0.00	
GST					0.00%	\$0.00	
Grand Total						\$500.00	

Users can reorder list items through drag-and-drop functionality, enabling personalized workflow customization.

Expenses

Expense EXP-00001

Pending

Created On 30/04/2025 09:00 AM

Created By John Doe

Filter

Sort by

Expense Name

Purchase of Glass Panels

Expense Date

01/05/2025

User

U-00002 - Adam Lim

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris accumsan facilisis mi at sodales. Integer congue mauris a mattis dictum. Nulla ac sem sit amet augue aliquet convallis ut ac odio. Aliquam viverra mi ut erat accumsan cursus. Cras id sollicitudin nisl. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet. Cras id sollicitudin nisl. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet.

Project

P-00001 - 45 Kallang Pudding Road, #07-12 Alpha Bui...

Supplier Name

Mr Glass Co.

Contact No.

+65 1000222

Attachment

quotation.pdf

Upload

Supplier Address

Jurong East 2, 08-2888, Business Park, S(998000)

Item	Description	Qty	U.m.	Unit Rate	Amount
1	Custom Glass Panels for Living Room	20	sq ft.	\$25.00	\$500.00
Subtotal					\$500.00
Discount				\$	\$0.00
GST				0.00%	\$0.00
Expense submitted successfully!					

A confirmation prompt will be displayed at the bottom right and an email notification is sent to the admin.

Project Overview

User Research

The Design

→ **Refinements**

Lessons Learned

REFINEMENTS

Usability Findings

① Better Contact Tracking

Users want a way to have a record of the clients to avoid constantly entering details of repeating clients.

② Confusion on the navigation bar

Ambiguous icons in the navigation bar lack intuitive meaning, leading to user confusion and incorrect assumptions about their functionality.

③ Lack flexibility for line items

Users need a bulk-removal feature for line items as manually deleting them one by one is time consuming and frustrating.

REFINEMENTS

Changes 1: Contacts

Before

Expense EXP-00001

Draft

SUBMIT

SAVE & CLOSE

Created On 30/04/2025 09:00 AM

Created By John Doe

Expense Name

Purchase of Glass Panels

Expense Date

01/05/2025

User

U-00002 - Adam Lim

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris accumsan facilisis mi at sodales. Integer congue mauris a mattis dictum. Nulla ac sem sit amet augue aliquet convallis ut ac odio. Aliquam viverra mi ut erat accumsan cursus. Cras id sollicitudin nisl. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet.

Project

P-00001 - 45 Kallang Pudding Road, #07-12 Alpha Bui...

Supplier Name

Mr Glass Co.

Contact No.

+65 1000222

Attachment

quotation.pdf

Upload

Supplier Address

Jurong East 2, 08-2888, Business Park, S(998000)

Item	Description	Qty	U.m.	Unit Rate	Amount
1	Custom Glass Panels for Living Room	20	sq.ft	\$25.00	\$500.00

After

Expense EXP-00001

Draft

SUBMIT

SAVE & CLOSE

Created On 30/04/2025 09:00 AM

Created By John Doe

Expense Name

Purchase of Glass Panels

User

U-00001 - John Doe

Project

P-00001 - 45 Kallang Pudding Road, #07-12 Alpha Bui...

Contact Name

Spaces Renovation Pte Ltd

Expense Date

01/05/2025

Due Date

30/05/2025

Contact Person

Glass & Aluminium Hub - Thomas Lee

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris accumsan facilisis mi at sodales. Integer congue mauris a mattis dictum. Nulla ac sem sit amet augue aliquet convallis ut ac odio. Aliquam viverra mi ut erat accumsan cursus. Cras id sollicitudin nisl. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet.

Address

45 Kallang Pudding Road, #07-12 Alpha Building, 349...

Attachment

quotation.pdf

Upload

The contact fields are all updated to use drop-down fields to allow ease of selection

Expense EXP-00001

Draft

SUBMIT

SAVE & CLOSE

Created On 30/04/2025 09:00 AM

Created By John Doe

Expense Name

Purchase of Glass Panels

User

U-00001 - John Doe

Project

P-00001 - 45 Kallang Pudding Road, #07-12 Alpha Bui...

Contact Name

Spaces Renovation Pte Ltd

Expense Date

01/05/2025

Due Date

30/05/2025

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris accumsan facilisis mi at sodales. Integer congue mauris a mattis dictum. Nulla ac sem sit amet augue aliquet convallis ut ac odio. Aliquam viverra mi ut erat accumsan cursus. Cras id sollicitudin nisl. Donec iaculis dolor eros, in imperdiet eros lacinia sit amet.

Contact Selection

Search...

SR Spaces Renovation Pte Ltd

MG Mr Glass Co. Pte Ltd

AD Aether Dynamics Pte Ltd

VI Vertex Innovations Pte Ltd

TS TerraGreen Solutions

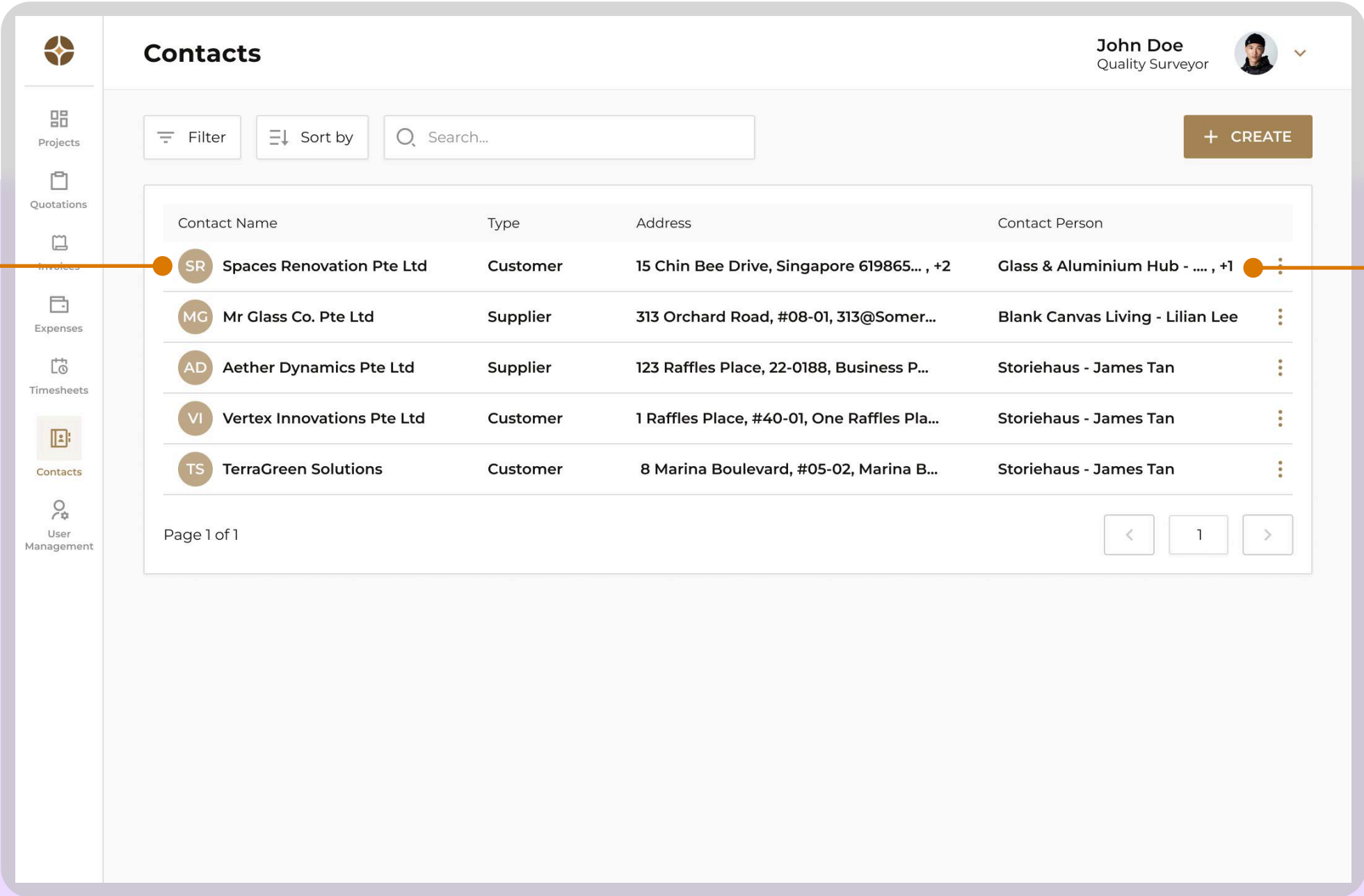
Attachment

quotation.pdf

Upload

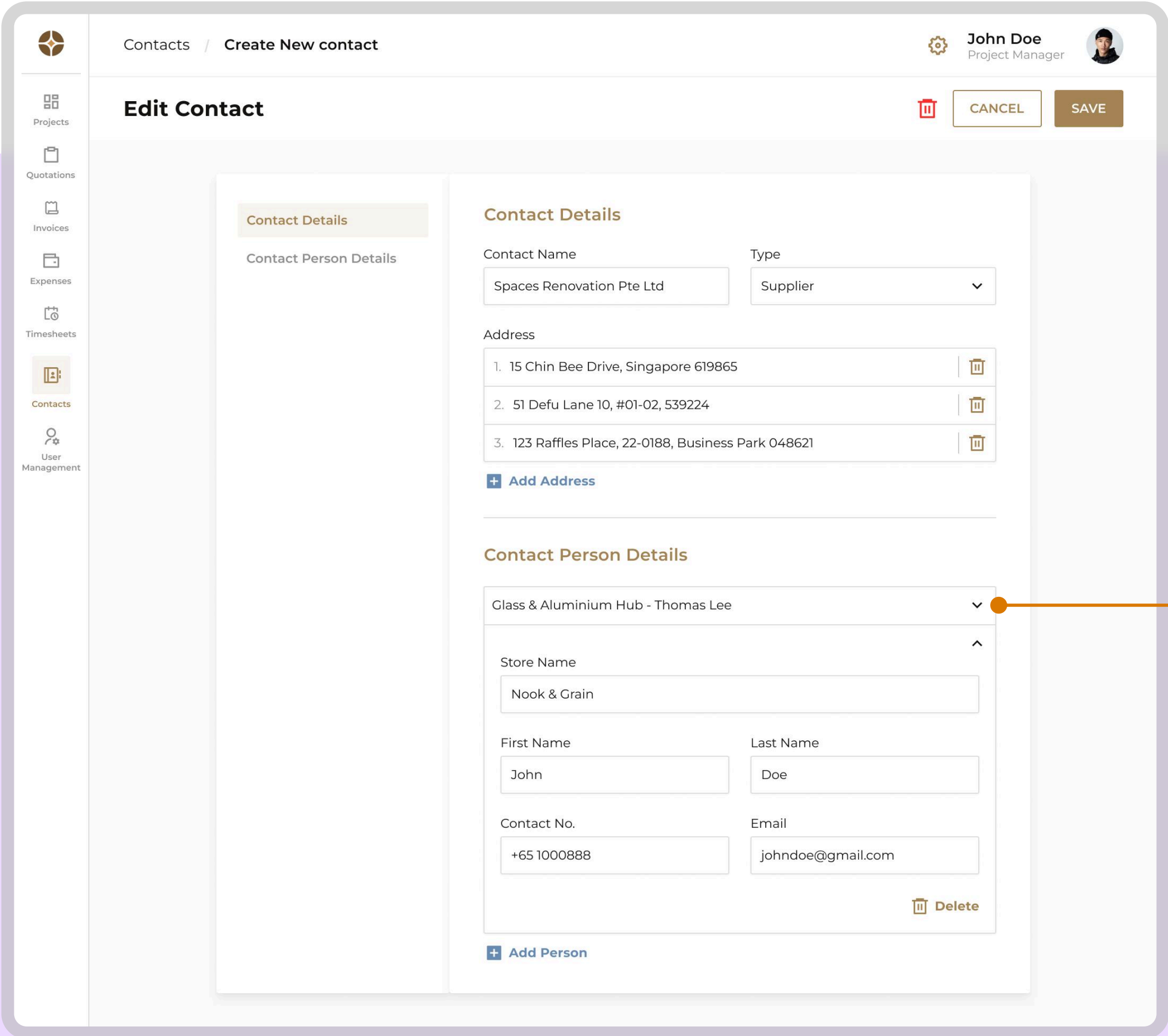
The contact selection dropdown includes integrated search functionality, enabling quick lookup without manual scrolling through lengthy lists.

Changes 1: Contacts



Using initial profile pictures establishes immediate visual hierarchy for contacts to appear as people rather than text entries.

A numeric counter appears when multiple contacts/addresses exist, giving visibility of the total count.



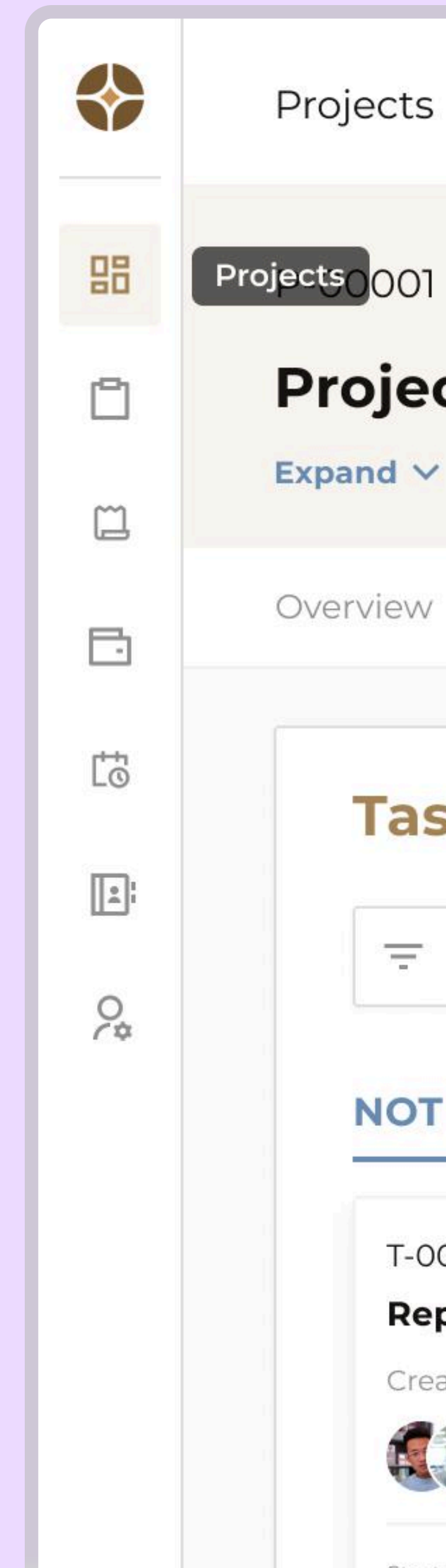
When the contact is collapsed only the store name and person-in-charge name is displayed to minimize the clutter.

REFINEMENTS

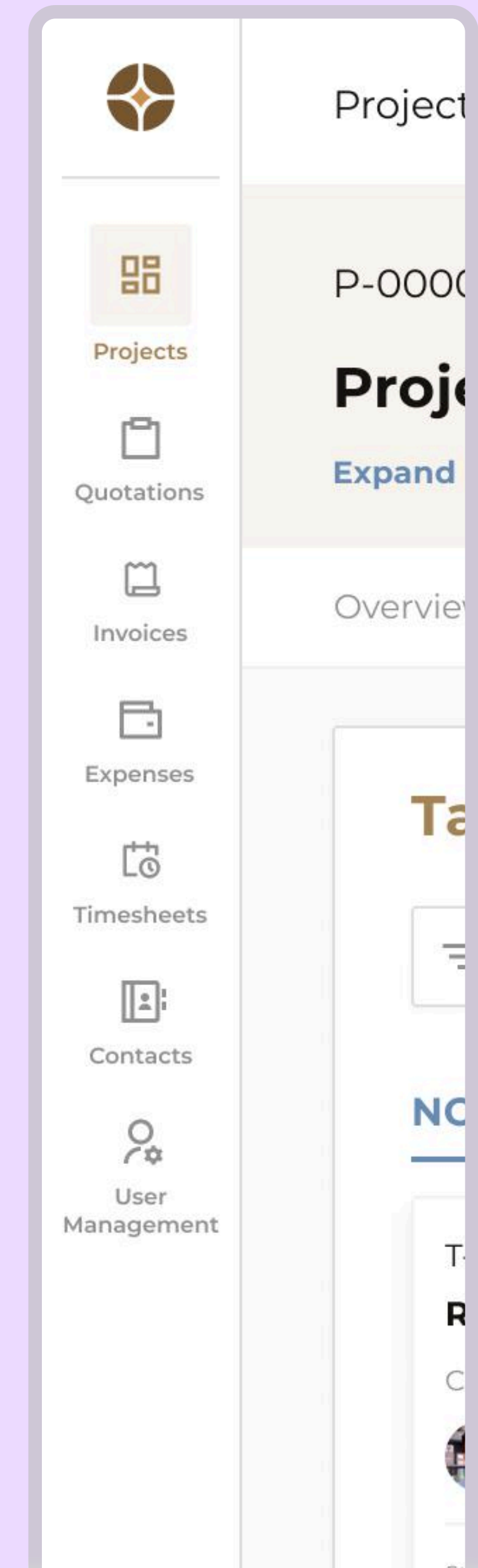
Changes 2: Navigation Bar

Based on stakeholder input that icons weren't self-explanatory and ambiguous. Hence, the navigation labels were changed from hover-reveal labels to permanently visible text labels for improve usability.

Before



After



Changes 3: Lack Flexibility

Before

Projects

Quotations

Invoices

Expenses

Timesheets

Contacts

User Management

Quotations / Quotation Q-00001

John Doe
Quality Surveyor

Quotation Q-00001

Draft

PRINT

Created On 30/04/2025 09:00 AM

Created By John Doe

Quotation Name

Project H Development

Type

Project

Currency

SGD

Quotation Date

01/05/2025

Valid Till

01/05/2025

QS in Charge

Adam Lim

PO Number

PO-00001

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris accumsan facilisis mi at sodales. Integer congue mauris a mattis dictum. Nulla ac sem sit amet augue aliquet convallis ut ac odio. Aliquam viverra mi ut erat accumsan cursus.

Client Name

Jane Lee

Client Contact

+65 1000888

Client Address

8 Boon Lay Way
#05-18 TradeHub 21
609964

Item	Description	Qty	U.m.	Excl. Amount	Remarks	Unit Rate	Amount
1	To remove window stickers inclusive of cleaning affec...	1	sets		Enter Remarks	\$1,000.00	\$1,000.00
Subtotal							\$1,000.00

Enter Project Details

View more

Grand Total

\$1,060.00

Require the deletion of line items to be done one by one

After

Projects

Quotations

Invoices

Expenses

Timesheets

Contacts

User Management

Quotations / Quotation Q-00001

John Doe
Quality Surveyor

Quotation Q-00001

Draft

PRINT

Created On 30/04/2025 09:00 AM

Created By John Doe

Quotation Name

Project H Development

Type

Project

Currency

SGD

Quotation Date

01/05/2025

Valid Till

01/05/2025

QS in Charge

Adam Lim

PO Number

PO-00001

Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris accumsan facilisis mi at sodales. Integer congue mauris a mattis dictum. Nulla ac sem sit amet augue aliquet convallis ut ac odio. Aliquam viverra mi ut erat accumsan cursus.

Client Name

Jane Lee

Client Contact

+65 1000888

Client Address

8 Boon Lay Way
#05-18 TradeHub 21
609964

3 Selected

Delete Items

1	To remove window stickers inclusive of cl...	1	sets		Enter Remarks	\$1,000.00	\$1,000.00
2	To remove window stickers inclusive of cl...	1	sets		Enter Remarks	\$1,000.00	\$1,000.00
3	To remove window stickers inclusive of cl...	1	sets		Enter Remarks	\$1,000.00	\$1,000.00
Subtotal							\$1,000.00

Enter Project Details

View more

Grand Total

\$1,060.00

Checkbox selection layout to allow for batch operations.

The header will display a real-time selection counter when items are selected.

Project Overview

User Research

The Design

Refinements

→ **Lessons learned**

Key Takeaways

Impact

The client expressed satisfaction with the initial deliverables, as the partially deployed system is already showing value in their internal operations.

What I have learned

This experience reinforced how critical it is to incorporate diverse user contexts and accessibility, while taking every feedback received as valuable perspective rather than criticism to enhance design outcomes.